0.1 Block-Jacobi normalization and the CG factor

Source items 7.4 and 7.6; module BlockJacobi

Work on a nonempty finite complex coordinate space with Euclidean operator norms. Assume D and $A = D + K$ are positive definite. Define

$$S = (\sqrt{D})^{-1}, \qquad B = SAS, \qquad E = SKS.$$

The square root is the actual positive matrix square root supplied by continuous functional calculus; the inverse is the actual matrix inverse. Positive definiteness makes $\sqrt{D}$ a unit. Its square identity and inverse identities prove

$$SDS = I, \qquad B = I + E.$$

A direct inverse bound. If $C = (I + E)^{-1}$ and $\|E\|_2 \le \eta < 1$, then $C = I - EC$. Consequently

$$(1 - \eta)\,\|C\|_2 \le 1, \qquad \|I + E\|_2 \le 1 + \eta.$$

The constructed normalization is invertible because S and A are. Combining the estimates gives the actual condition number bound

$$\boxed{\kappa_2(B) := \|B\|_2 \left\|B^{-1}\right\|_2 \le \frac{1+\eta}{1-\eta}.}$$

This norm argument does not require a separately assumed Hermitian perturbation. Nonnegative η follows from the norm bound. Nonemptiness is explicit because the condition-number and norm-of-identity convention uses a nonzero coordinate space.

The preconditioning corollary. Define

$$q(\kappa) = \frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1}.$$

Monotonicity of square root and positive denominators prove that q is monotone. Lean also proves $\kappa_2(B) \ge 1$, $q(\kappa) \ge 0$ for $\kappa \ge 1$, and $q(\kappa) < 1$. Thus

$$\boxed{q(\kappa_2(B)) \le q\left(\frac{1+\eta}{1-\eta}\right).}$$

For $1 \le \kappa$ and $e_0 \ge 0$, an explicit classical premise $e_m \le 2q(\kappa)^m e_0$ and a certified upper bound $\kappa \le \bar{\kappa}$ yield $e_m \le 2q(\bar{\kappa})^m e_0$.

What is not claimed. The classical conjugate-gradient convergence theorem, item 7.5, is not newly formalized. The new corollary proves its factor comparison and bound substitution. It does not replace the original theorem's algorithm hypotheses with a new axiom or claim a floating-point CG convergence proof.